# Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

**Ringkasan**

PENDAHULUAN
TINJAUAN PUSTAKA
[T1]fontenc [utf8]inputenc fourier tgheros [cmex10]amsmath amssymb [pdftex]graphicx epstopdf
fancyhdr [twoside, margin=2.5cm, bindingoffset=0.0cm]geometry [numbers,sortcompress]natbib url
siunitx detect-all,input-product=*[labelformat=simple]subcaption font=footnotesize,sf,skip=2pt [sub]labelfont=scriptsiz
[sectcntreset]bibtopic [compress,capitalize]cleveref blindtext microtype
L>$l$ <
firstpage [R] [L]DOCTORAL PROGRAM IN
MICROSYSTEMS & MICROELECTRONICS

**Annual Thesis Report (2013–2014)**

## Modeling nanoscale quasi-ballistic MOS transistors:

## a circuit design perspective

| | |
|---|---|
| **Report Type:** | Final |
| **PhD Student:** | Anurag Mangla |
| **Thesis Director:** | Prof. Christian Enz |
| **Thesis Co-director:** | Dr. Jean-Michel Sallese |
| **Date:** | 19 Pebruari 2025 |

Anurag Mangla

0.245

Gambar 1:
0.245

Gambar 2:
0.245

Gambar 3:
0.245

Gambar 4:

Gambar 5: fig:1a,fig:1c Potential profile fig:1b,fig:1d Carrier density profile of the 10;100 ballistic and diffusive devices. $V_{\mathrm{D}}$ =0.5V, $V_{\mathrm{S}}$ =0V.

Gambar 6: Geometry of the proposed partially gated DGSOI MOSFET. The channel length $L_{\mathrm{c}}$ is the length of the semiconductor between the source and drain junctions. The gate length $L_{\mathrm{G}}$ is the length of the channel covered by the metal gate. This structure was simulated for $L_{\mathrm{G}}/L_{\mathrm{c}}$ = 0.1;0.3;0.5;0.9;1.



# 1 Work Objectives

The goals of this work are as following:

- Understanding the behavior of ballisticity in nanoscale MOSFETs focusing especially on the electrostatics.
- Modeling the charge and potential in ballistic and quasi-ballistic double-gate MOSFETs in a semi-empirical manner that is more approachable to a designer.

# 2 Work Achieved in the Past Year

## 2.1 Subsection heading 1

## 2.2 Subsection heading 2

## 2.3 Subsection heading 3

# 3 Current State of the Work

Writing and editing of the PhD dissertation has begun and is expected to be finished by the end of June 2014.

# 4 Calendar of Upcoming Work

| | |
|---|---|
| Submission of the thesis | June 2014 |
| Private thesis defense | August 2014 |

[IEEEtranN]AnnualReport2014

# 5 Scientific Publications, Presentations, Conferences

[IEEEtran]BallisticVirtualSource

# 6 References

0.245

Gambar 7:
0.245

Gambar 8:
0.245

Gambar 9:
0.245

Gambar 10:
0.245

Gambar 11:
0.245

Gambar 12:
0.245

Gambar 13:
0.245

Gambar 14:

Gambar 15: fig:3:2a,fig:3:4a,fig:3:3a,fig:3:5a Potential profile fig:3:2b,fig:3:4b,fig:3:3b,fig:3:5b Carrier density profile of the 10;100 partial-gate devices with different $L_G/L_c$ ratios. $V_D$ =0.5V, $V_S$ =0V